

CCNA

VLAN and INTER-VLAN Routing TASK – 3

1. In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

Ans :

Internal Network

External Network

Internal PC

External PC

192.168.1.10

203.0.113.10

```
|
|
+---- G0/0 [Router] G0/1 -----+
      192.168.1.1    203.0.113.1
```

- enable
- configure terminal

- interface gigabitEthernet 0/0
- ip address 192.168.1.1 255.255.255.0
- ip nat inside
- no shutdown
- exit

- interface gigabitEthernet 0/1
- ip address 203.0.113.1 255.255.255.0
- ip nat outside
- no shutdown
- exit
- ip nat inside source static 192.168.1.10 203.0.113.5
- end
- show ip nat translations

**2. Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.**

Ans :

- enable
- configure terminal
- interface gigabitEthernet 0/0
- ip address 192.168.1.1 255.255.255.0
- ip nat inside
- no shutdown
- exit
- interface gigabitEthernet 0/1
- ip address 203.0.113.1 255.255.255.0

- ip nat outside
- no shutdown
- exit
- access-list 1 permit 192.168.1.0 0.0.0.255
- ip nat inside source list 1 interface gigabitEthernet 0/1 overload
- end
- show ip nat translations
- **PC1:** 192.168.1.10 / 255.255.255.0 / Gateway 192.168.1.1
- **PC2:** 192.168.1.11 / 255.255.255.0 / Gateway 192.168.1.1
- **Web Server:** 203.0.113.10 / 255.255.255.0 / Gateway 203.0.113.1

3. Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

Ans :

- enable
 - show ip nat translations
 - show ip nat statistics
- 4. Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.**

Ans :

- enable
- configure terminal
- interface gigabitEthernet 0/0
- ip address 192.168.1.1 255.255.255.0
- ip nat inside
- no shutdown
- exit
- interface gigabitEthernet 0/1
- ip address 203.0.113.1 255.255.255.0
- ip nat outside
- no shutdown
- exit
- access-list 1 permit 192.168.1.0 0.0.0.255
- ip nat inside source list 1 interface gigabitEthernet 0/1 overload
- end
- show ip nat translations
- show ip nat statistics

- PC1: 192.168.1.10 / 255.255.255.0 / Gateway 192.168.1.1
- PC2: 192.168.1.11 / 255.255.255.0 / Gateway 192.168.1.1
- PC3: 192.168.1.12 / 255.255.255.0 / Gateway 192.168.1.1

5. Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

Ans : NAT (Network Address Translation) allows private IP addresses inside a network to communicate with the Internet by translating them into public IP addresses. PAT (Port Address Translation) goes further by allowing many devices to share one public IP address using different port numbers. This helps large companies such as Zomato or Flipkart conserve their limited public IPv4 addresses while supporting millions of connections. For example, thousands of employees or internal servers can access external services through a smaller pool of public addresses. PAT keeps each connection separate by tracking the source port, destination, and IP information. As a result, organizations can scale their networks without needing a unique public IPv4 address for every device.